

AI & Machine Learning for Civil Engineers

From engineering data to prediction, explainability and uncertainty, in eight hands-on sessions.

08

Live sessions

90 min

Per session

04 weeks

Course duration

**Runs 100% on Google Colab**

Nothing to install, works even on a Chromebook

**Built for beginners**

No prior Python or machine learning required

**Live code-along on Civil Engineering datasets**

Materials · Buildings & Energy · Transport · Environment · Geotechnics

**Hands-on notebooks, cross-domain assignments & solutions**

Plus a certificate and session recordings*

* Recordings shared after each session; available until two weeks after the course ends.

Course outline

- 01 Python for Civil Engineers**
AI/ML basics, Colab, NumPy & pandas with engineering data
- 02 From Raw Data to ML-Ready**
Explore, clean, visualize & split engineering data
- 03 Your First Prediction Model**
Linear regression, error metrics & cross-validation
- 04 Trees & Ensemble Models**
Decision Trees, Random Forest & boosting for nonlinear problems

Designed for Civil Engineers across disciplines — Materials, Structures, Geotechnics, Transportation, Water & Environment.

INSTRUCTOR

Dr. Abdul Waris

Research Associate

Universität Siegen (University of Siegen)

Former Prime Minister's Research Fellow

Ph.D. in Geotechnical Engineering, IIT Hyderabad

- Research Excellence Award 2026, IIT Hyderabad
- Invited talks at Oregon State University (USA) and a Faculty Development Programme at D.Y. Patil (Pune, India)
- Invited resource person, five-day ML workshop at NIT Srinagar

ONE ML WORKFLOW. MULTIPLE CIVIL ENGINEERING APPLICATIONS.

Problem → Data → Model → Validate → Interpret → Decide

- 05 Classification for Engineering Decisions**
Logistic regression, confusion matrix, precision & recall
- 06 Artificial Neural Networks**
Architecture, activations, training & overfitting
- 07 Explainable AI & Engineering Judgement**
Feature importance, SHAP, and engineering interpretation
- 08 From Prediction to Probability**
Monte Carlo, ML surrogates & probability of failure

WHO SHOULD JOIN

B.Tech & M.Tech students / PhD scholars / Faculty
Practising engineers / Complete beginners to AI & ML

STARTS · LIMITED SEATS

**Saturday,
17 October 2026**Scan to register · qudeltastudios.com →